

Relational Databases

COM 736 Database Systems and Data Analytics



Objectives

Database Models

- Brief Overview

Relational Model

- History and data storage

Database

- A Database is an **integrated collection** of stored operational data used by the application systems of a particular enterprise.
 - Data for all applications in the enterprise is stored in the integrated database **not** individual files.
 - **Centralised control** of its operational data
 - **Data independence**

Database Models

Hierarchical Model and Network Model

- Early models, closely associated with physical storage and programs are written to interact with the data

Relational Model

- Conceptual model and physical storage and separated with a Structured Query Language to interact through DMBS

Object Oriented Model

- Dealing with complex objects using object oriented model

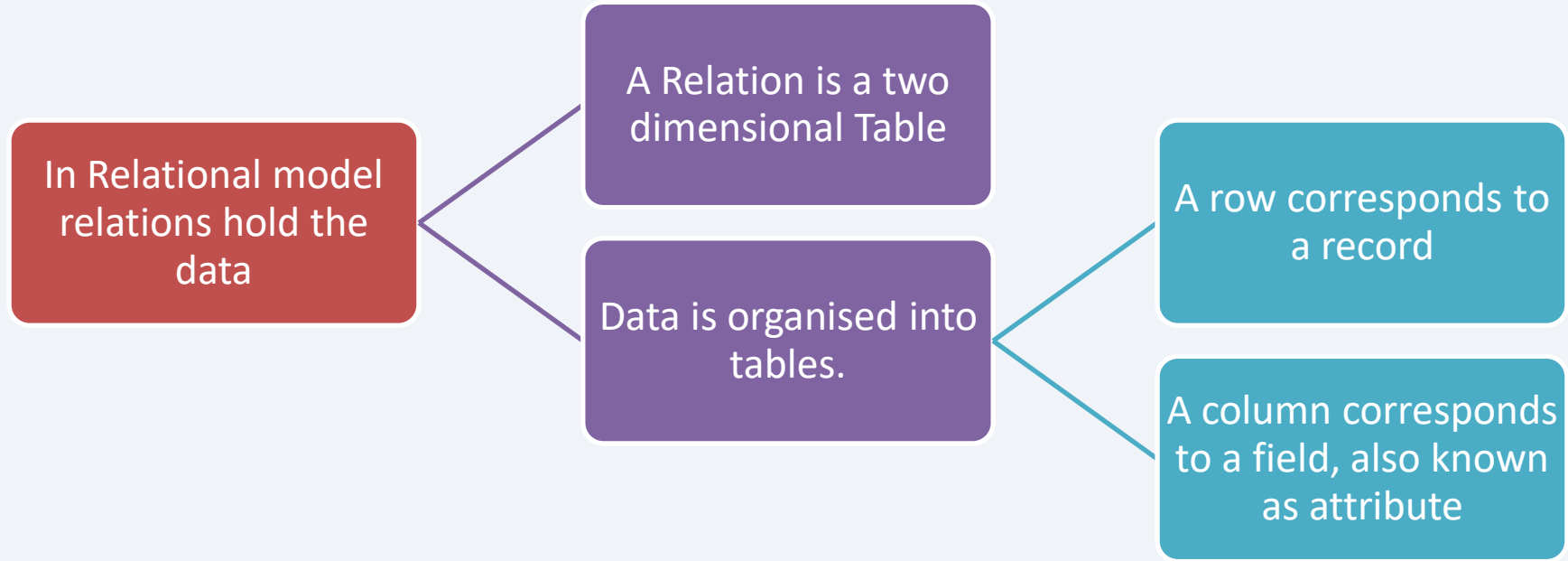
XML Document Model

- eXtended Mark-up Language for web applications based on document models and relational models

Big Data Systems and NoSQL Database Model

- Fulfilling the requirements of non traditional data type storage and retrieval, fast access of enormous data through various applications and platforms.

The Relational Model



The Relational Model

Tables are different from sequential files

Each table contains only one type of record

Each record in a table has the same number of fields

The order of the records within a table has no significance

The Relational Model

Customer : Table

Attributes

Records

CustomerNumber	Name	Address	Type
5567	Jones	Cross Rd	Business
2913	Brown	Cross Rd	Personal
4890	Jones	Main St	Business

The Relational Model

- Each row of a table is **uniquely identified** by the data values from one or more columns.
 - Known as **Primary Key**
 - In this customer table, Number is the primary key

CustomerNumber	Name	Address	Type
5567	Jones	Cross Rd	Business
2913	Brown	Cross Rd	Personal
4890	Jones	Main St	Business

- If there are more than one possible way of uniquely identifying each record of a table, each known as a Candidate key.
 - One of the candidate keys must be selected as the primary key

The Relational Model

- The relationships between rows of data in different tables are represented solely by the data values of one or more column.

COURSE

Course_ID	Course_Name	Level
BSC_COMP	BSc Computing	Undergraduate
MSC_COM_SCI	MSc Computer Science	Postgraduate
MSC_COM_NW	MSc Computer Science with Networking	Postgraduate
MSC_COM_BDA	MSc Computer Science with Big Data Analytics	Postgraduate

STUDENT

Student_ID	First_Name	Last_Name	Course_ID
S001	Jack	Adams	BSC_COMP
S002	Jill	Jones	MSC_COM_SCI
S003	Bob	Who	MSC_COM_SCI
S004	Alice	Davies	MSC_COM_BDA

The Relational Model

- A field referring to the primary key field of another table is a foreign key.

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Primary key – Course_ID

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S003	Bob	Who	MSC_COM_SCI
S004	Alice	Davies	MSC_COM_BDA

Primary Key – Student_ID

Foreign key – COURSE_ID

Summary

- Database
- Database Models
- Relational Model
- Next:
 - Database Design: Entity Relationship modelling